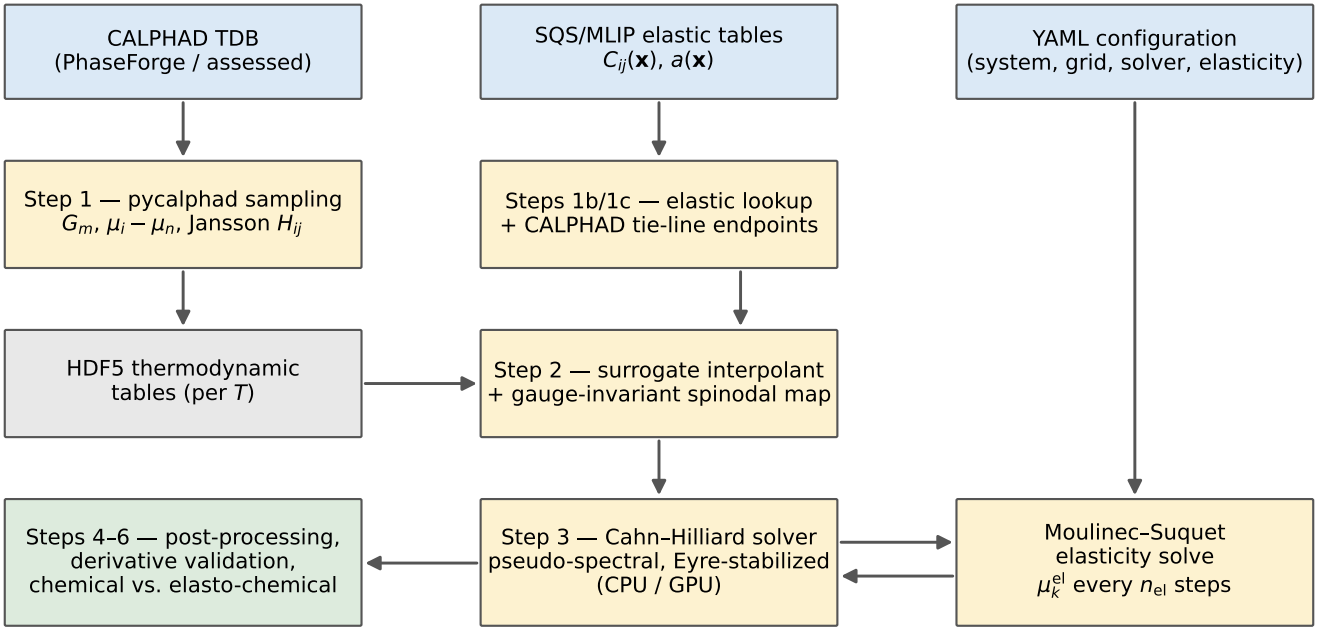
CALPHAD TDB
(PhaseForge / assessed)
SQS/MLIP elastic tables
$C_{ij}(\mathbf{x})$, $a(\mathbf{x})$
YAML configuration
(system, grid, solver, elasticity)
Step 1 — pycalphad sampling
G_m, $\mu_i - \mu_n$, Jansson H_{ij}
Steps 1b/1c — elastic lookup
+ CALPHAD tie-line endpoints
HDF5 thermodynamic
tables (per T)
Step 2 — surrogate interpolant
+ gauge-invariant spinodal map
Steps 4–6 — post-processing,
derivative validation,
chemical vs. elasto-chemical
Step 3 — Cahn–Hilliard solver
pseudo-spectral, Eyre-stabilized
(CPU / GPU)
Moulinec–Suquet
elasticity solve
μ_k^{el} every n_{el} steps